Model Summary Table

Component Layer	Input Shape	Output Shape	Total Parameters	Trainable Status (Stage 1 / Stage 2)
InputLayer	$(224, 224, 3)$	$(224, 224, 3)$	0	Non-trainable
Data Augmentation	$(224, 224, 3)$	$(224, 224, 3)$	0	Non-trainable (GPU execution)
EfficientNetB0 Backbone	$(224, 224, 3)$	$(7, 7, 1280)$	4,049,571	Frozen / Top 20 Layers Unfrozen
GlobalAveragePooling2D	$(7, 7, 1280)$	(1280)	0	Non-trainable
Dropout (0.2)	(1280)	(1280)	0	Non-trainable
Dense Output Head	(1280)	(10)	12,810	Trainable / Trainable

Visual Architecture Flow Diagram

[Input Image Tensor: 224 x 224 x 3]

[Dynamic GPU Data Augmentation] (Flip, Rotation, Zoom, Translation)

[EfficientNetB0 Base Backbone] (237 Conv & MBConv Block Layers)

▼ Output Feature Map: 7 x 7 x 1280 (62,720 values)

[GlobalAveragePooling2D Layer] (Spatial Mean Reduction)

▼ Compressed Feature Vector: 1280 values

[Dropout Regularization Layer] (Deactivates random 20% during training)

▼ Regularized Vector: 1280 values

[Dense Softmax Output Layer] (10 Class Probability Units)

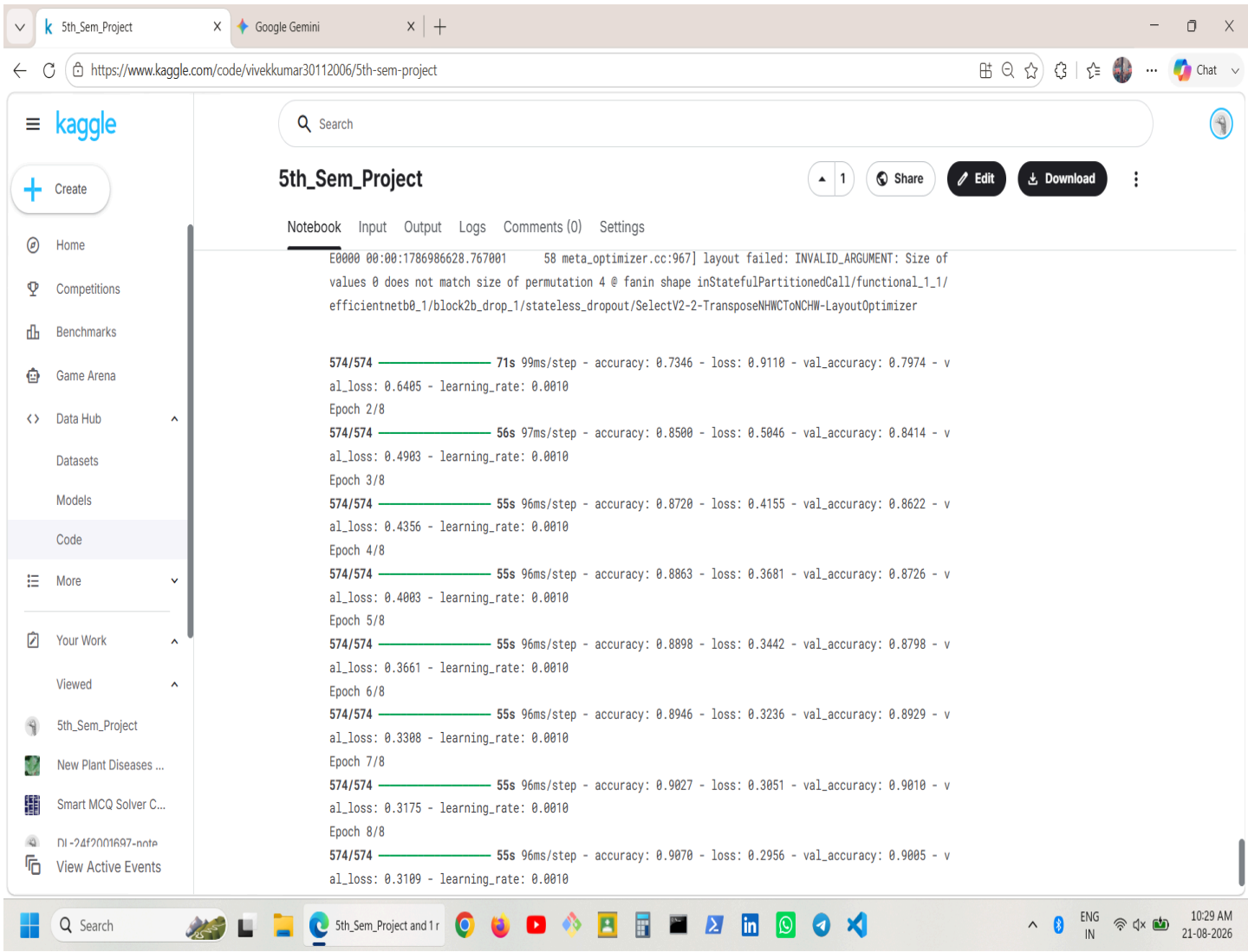
Model Background & Technical Summary

- **Developer & Release:** Developed by the Google Brain team (Mingxing Tan & Quoc V. Le) in 2019.
- **Original Pre-Training Dataset:** Pre-trained on the **ImageNet (ILSVRC)** dataset, comprising **1.4 million annotated images** categorized into **1,000 general object classes**.
- **Architecture Discovery:** Built using **Neural Architecture Search (NAS)** to discover the optimal network structure that maximizes classification accuracy while minimizing floating-point operations (FLOPs).

Layer Breakdown

- **Base Model (EfficientNetB0): 237 internal layers** (including standard convolutions, depthwise separable convolutions, batch normalization steps, Swish activation functions, and Squeeze-and-Excitation attention blocks).
- **Full Pipeline Layers: 241 total layers** when combined with custom operational stages:
    1. InputLayer (Receives $224 \times 224 \times 3$ image tensors)
    2. Data Augmentation (GPU-based spatial transformations)
    3. EfficientNetB0 Backbone (237 feature extraction layers)
    4. GlobalAveragePooling2D (Spatial feature compression)
    5. Dropout(0.2) (Regularization layer)
    6. Dense (10-class Softmax classification head)

WITHOUT FINE TUNING EfficientNetB0 MODEL TRAINING:



5th_Sem_Project | Google Gemini

https://www.kaggle.com/code/vivekkumar30112006/5th-sem-project

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```
Starting Stage 2 Fine-Tuning...
Epoch 1/8


E0000 00:00:1786987087.894222      58 meta_optimizer.cc:967] layout failed: INVALID_ARGUMENT: Size of values 0 does not match size of permutation 4 @ fanin shape inStatefulPartitionedCall/functional_1_1/efficientnetb0_1/block2b_drop_1/stateless_dropout/SelectV2-2-TransposeNHWCToNCHW-LayoutOptimizer


574/574 ━━━━━━━━━━━━━━━━━━━━ 78s 112ms/step - accuracy: 0.8096 - loss: 0.5653 - val_accuracy: 0.8674 - val_loss: 0.3985 - learning_rate: 1.0000e-05
Epoch 2/8
574/574 ━━━━━━━━━━━━━━━━━━━━ 61s 107ms/step - accuracy: 0.8682 - loss: 0.3978 - val_accuracy: 0.8846 - val_loss: 0.3479 - learning_rate: 1.0000e-05
Epoch 3/8
574/574 ━━━━━━━━━━━━━━━━━━━━ 61s 106ms/step - accuracy: 0.8952 - loss: 0.3178 - val_accuracy: 0.8947 - val_loss: 0.3111 - learning_rate: 1.0000e-05
Epoch 4/8
574/574 ━━━━━━━━━━━━━━━━━━━━ 61s 107ms/step - accuracy: 0.9023 - loss: 0.2877 - val_accuracy: 0.9073 - val_loss: 0.2750 - learning_rate: 1.0000e-05
Epoch 5/8
574/574 ━━━━━━━━━━━━━━━━━━━━ 62s 108ms/step - accuracy: 0.9175 - loss: 0.2519 - val_accuracy: 0.9156 - val_loss: 0.2519 - learning_rate: 1.0000e-05
Epoch 6/8
574/574 ━━━━━━━━━━━━━━━━━━━━ 62s 107ms/step - accuracy: 0.9219 - loss: 0.2342 - val_accuracy: 0.9191 - val_loss: 0.2369 - learning_rate: 1.0000e-05
Epoch 7/8
574/574 ━━━━━━━━━━━━━━━━━━━━ 62s 108ms/step - accuracy: 0.9257 - loss: 0.2209 - val_accuracy: 0.9252 - val_loss: 0.2164 - learning_rate: 1.0000e-05
Epoch 8/8
574/574 ━━━━━━━━━━━━━━━━━━━━ 62s 107ms/step - accuracy: 0.9345 - loss: 0.2018 - val_accuracy: 0.9313 - val_loss: 0.1978 - learning_rate: 1.0000e-05
```

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TEST IMAGE PREDICTION

